

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement 1.2V NCAP Monte Carlo SPICE Model Document

**Version 0.5 Phase1
2011/08/01**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2 _p1	2010/06/28	Yan Zung	1. This Monte Carlo model includes both process and mis-matching variation characteristics extracted from model cards as below : L110AE_NCAP12_V021.mdl L110AE_NCAP12_V021.lib
0.3 _p1	2010/10/19	Yan Zung	1. Modified the model cards based on l110ae_ncap12_v031.mdl l110ae_ncap12_v031.lib (Removed the all description about m and ms_1 in the model cards, and revised the formula of Igate.)
0.4 _p1	2010/11/01	Laney Fan	1. Remove the header of the model 2. Modify the nf=1 to nf=2
0.5 _p1	2011/08/01	Laney Fan	1. Correct the include file name in the l110ae_ncap12_v051_mc.lib

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI Advanced Enhancement Process

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 1%. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.09
Cadence SPECTRE version 6.2.1.299
Mentor Graphics ELDO version 6.11_1.1

1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. The user can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo NCAP model
TT: Typical NCAP case
FF: Maximum NCAP case
SS: Minimum NCAP case

1.3.3 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

1.3.4 Model usage

For Synopsys HSPICE

-Copy model files into your simulation directory.

-Define sigma value for MC simulation.

Example: `.param sigma=3`

-Add the following statement in the input file to the simulator.

`.lib "/ModelFileName_mc_corner.lib" CaseName`

, where CaseName could be MC, TT, FF, SS.

-Define device condition by "Xxx" statement.

Example: `Xncap Vgg Vbb DeviceName lf=2u wf=5u nf=4 mis_flag=1`

-mis_flag: the flag parameter is used to turn on/off a mismatching parameter for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

-Define the control of Monte-Carlo simulation.

Indicator:

PROCESS: 1 to turn on the MC process simulation

0 to turn off the MC process simulation

MISMATCH: 1 to turn on the MC mismatch simulation

0 to turn off the MC mismatch simulation

Example 1:

`.PARAM PROCESS=1 MISMATCH=1`

* turn on MC process and mismatch simulations at the same time.

Example 2:

`.PARAM PROCESS=1 MISMATCH=0`

* just turn on the MC process simulation only.

Example 3:

`.PARAM PROCESS=0 MISMATCH=1`

* just turn on the MC mismatch simulation only.

- Define the number of Monte Carlo iterations to run MC simulations.

Example: `.AC LIN 2 10K 100K Sweep Monte=200`

For Cadence SPECTRE

-Copy model files into your simulation directory.

-Define sigma value for MC simulation.

Example: `.param sigma=3`

-Add the following statement in the input file to the simulator.
include "./ModelFileName_mc_corner.lib.scs" section=CaseName
, where CaseName could be mc, tt, ff, ss.

-Define device condition by "Xxx" statement.

Example: Xncap (Vgg Vbb) DeviceName lf=2u wf=5u nf=4 mis_flag=1

- mis_flag: the flag parameter is used to turn on/off a mismatching parameter for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

- Define the number of Monte Carlo iterations and control MC simulations.

Example 1: run mismatch simulation only

mc1 montecarlo numruns=200 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes

Example 2: run process simulation only

mc1 montecarlo numruns=200 seed=1 variations=process donominal=yes
savefamilyplots=yes

Example 3: run process & mismatch simulation together

mc1 montecarlo numruns=200 seed=1 variations=all donominal=yes
savefamilyplots=yes

For Mentor Graphics ELDO

-Copy following files into your simulation directory.

-Define sigma value for MC simulation.

Example: .param sigma=3

-Add the following statement in the input file to the simulator.

.lib "./ModelFileName_MC_CORNERlib.eldo" CaseName
, where CaseName could be MC, TT, FF, SS.

-Define device condition by "Xxx" statement.

Example: Xncap Vgg Vbb DeviceName lf=2u wf=5u nf=4 mis_flag=1

-mis_flag: the flag parameter is used to turn on/off a mismatching parameter for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.

- Define the control of Monte-Carlo simulation.

Indicator:

PROCESS: 1 to turn on the MC process simulation

0 to turn off the MC process simulation

MISMATCH: 1 to turn on the MC mismatch simulation

0 to turn off the MC mismatch simulation

Example 1:

PARAM PROCESS=1 MISMATCH=1

* turn on MC process and mismatch simulations at the same time.

Example 2:

PARAM PROCESS=1 MISMATCH=0

* just turn on the MC process simulation only.

Example 3:

.PARAM PROCESS=0 MISMATCH=1

* just turn on the MC mismatch simulation only.

- Define the number of Monte Carlo iterations to run MC simulations.

Example: .AC LIN 2 10K 100K

.MC 200

ModelFileName: l110ae_ncap12_v051_mc

DeviceName: ncap_12

The NCAP model is valid within the following geometry, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed. Due to the consideration of capacitance measurement resolution, this model card is suggested not to extrapolate to the devices with the area ($L_f \cdot W_f \cdot N_f \cdot M$) smaller than $2 \mu\text{m}^2$.

Geometry description (before 90% shrinkage):

$0.15 \mu\text{m} \leq L_f \leq 2 \mu\text{m}$

$2.5 \mu\text{m} \leq W_f \leq 10 \mu\text{m}$

$2 \leq N_f \leq 8$

Bias Range:

$-1.2\text{V} (*1.1) \leq V_{GS} \leq 1.2\text{V} (*1.1)$

$V_{DS} = V_{BS} = 0\text{V}$

Temperature range:

$-50^\circ\text{C} \sim +125^\circ\text{C}$

Layout example:

The following figure is the layout example of the NCAP. LF and WF are the single poly finger length and width, respectively, and Nf is the total finger number.

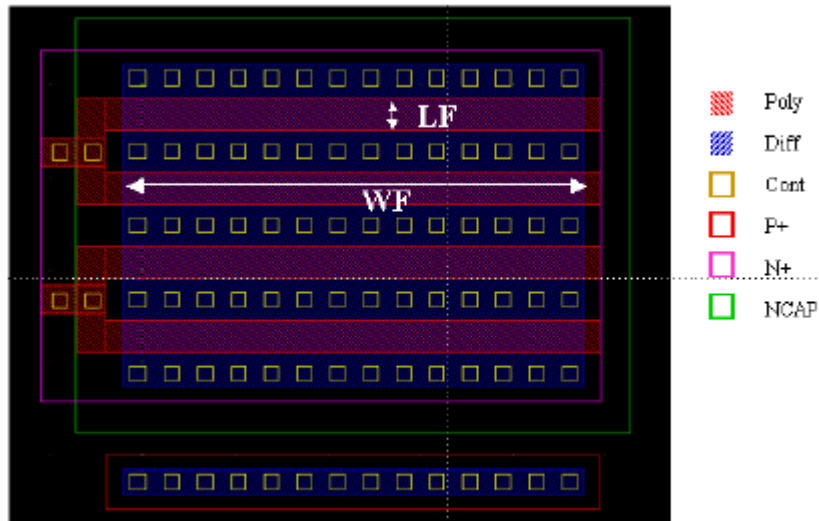


Figure 1-1 NCAP layout

2 Monte Carlo DC Simulation Results for Process Variation

Two hundred (200) Monte Carlo iterations are performed in the following simulation conditions and the following corner points show TT, FF, and SS values, individually.

A reasonable number of Monte Carlo iterations is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to $(\text{iteration times})^{(-1/2)}$.

2.1 Simulation Result Plots

The following figure show out NCAP's capacitance process variation at 25 °C :

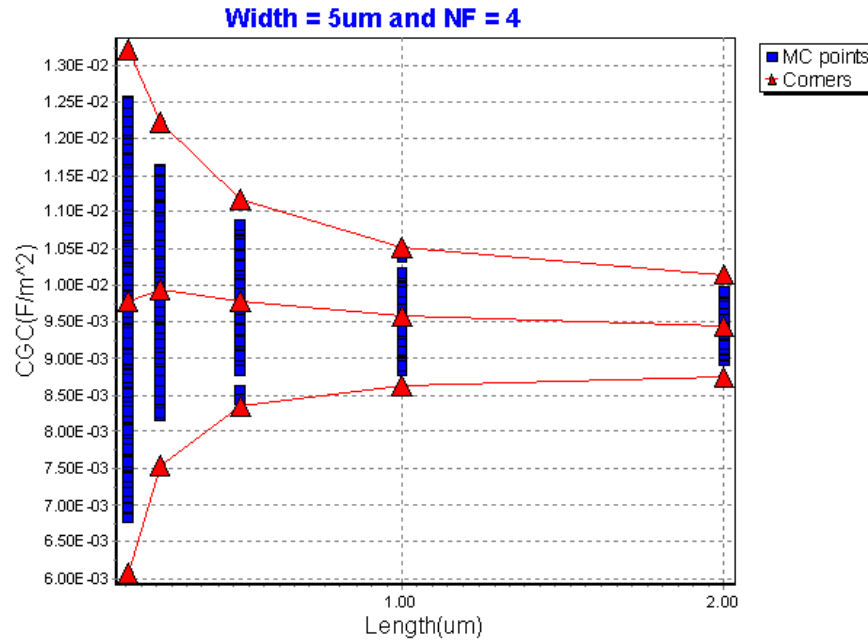


Figure 2-1 Monte Carlo plot of NCAP's capacitance.

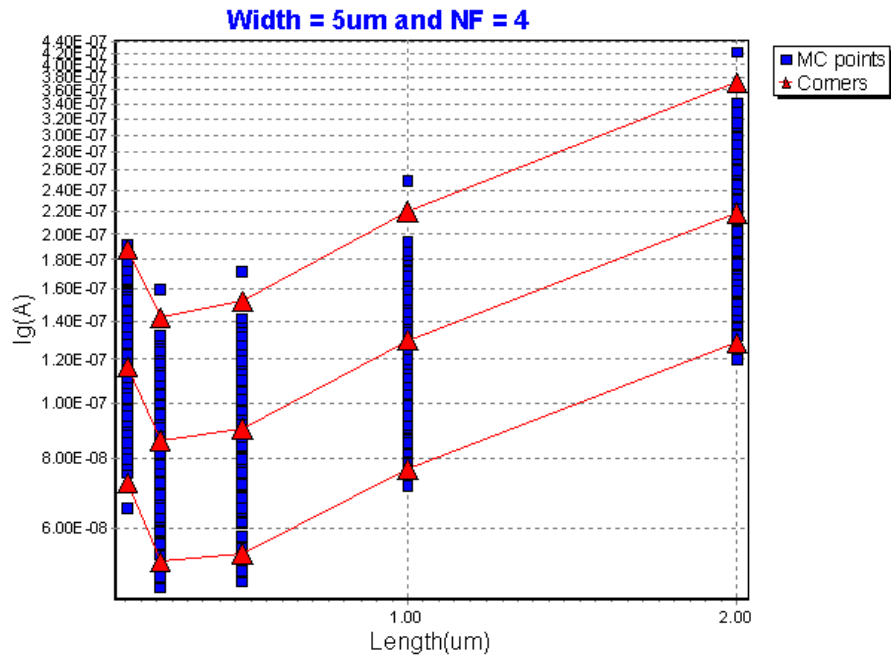


Figure 2-2 Monte Carlo plot of NCAP's gate current

2.2 Simulation Result Plot For Mismatching Variation

The following figure show out NCAP's capacitance mismatching variation at 25 °C :

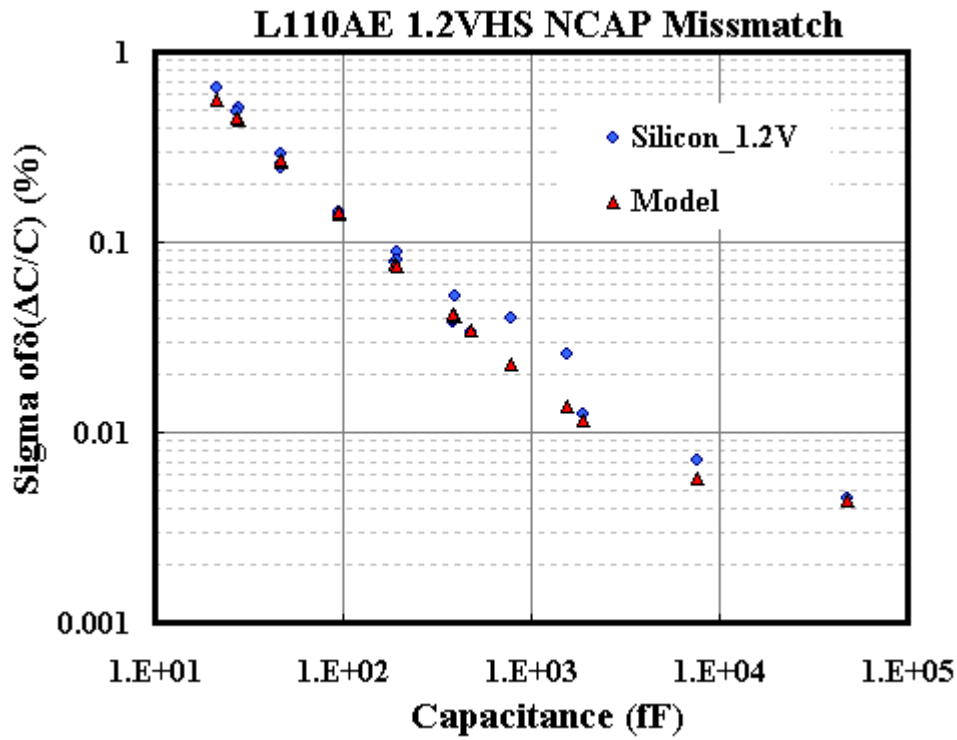


Figure 2-3 Mismatching simulation of NCAP's capacitance.

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1.	None	None